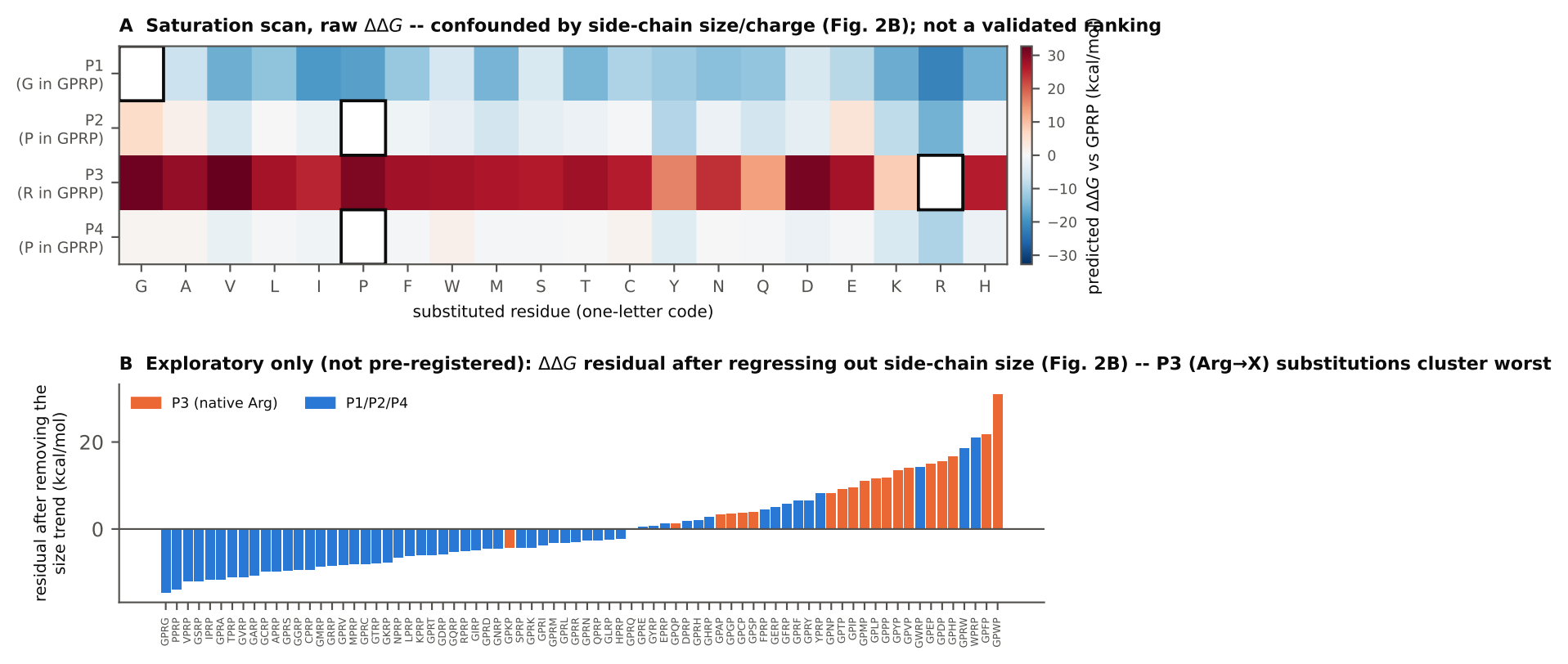
A Saturation scan, raw ΔΔG -- confounded by side-chain size/charge (Fig. 2B); not a validated ranking
P1 (G in GPRP)
P2 (P in GPRP)
P3 (R in GPRP)
P4 (P in GPRP)
G A V L I P F W M S T C Y N Q D E K R H
substituted residue (one-letter code)
predicted ΔΔG vs GPRP (kcal/mol)
30 20 10 0 −10 −20 −30
B Exploratory only (not pre-registered): ΔΔG residual after regressing out side-chain size (Fig. 2B) -- P3 (Arg→X) substitutions cluster worst
P3 (native Arg)
P1/P2/P4
residual after removing the size trend (kcal/mol)
20
0
GPRG PPRP VPRP GSRP IPRP GPRA TPRP GVRP GARP GCRP APRP GPRS GTRP GSRP GMRP GRRP GPRV MPRP GPRC GTRP GKRP NPRP LPRP KPRP GPRT GDRP GQRP RPRP GIRP GPRD GNRP GPKP SPRP GPRK GPRI GPRM GPRL GPRR GPRN QPRP GLRP HPRP GPRQ GPRE GPRF EPRP GPQP DPRP GPRH GHRP GPAP GPGP GPCP GPSP FPRP GERP GFRP GPRF GPRY YPRP GPNP GPTP GPIP GPMP GPLP GPPP GPYP GPVP GWRP GPEP GPDP GPHP GPRW WPRP GPFP GPWP